JIM SCHNEIDER: OK great. Good morning, everybody. Welcome to the final day of the Goldman Sachs Communicopia Technology Conference. My is Jim Schneider. I am the Semiconductor Analyst here at Goldman Sachs. It's my pleasure to welcome AMD to the stage today.

With us from the company we have SVP and General Manager of the Compute Enterprise AI, Dan McNamara, and Corporate Vice President of Finance, Strategy and Investor Relations, Matt Ramsay. Welcome, guys. Thanks for being here.

DAN MCNAMARA: Thank you.

MATT RAMSAY: Thank you.

JIM SCHNEIDER: I think the topic, almost every session at this conference, is AI. So you're a key enabler of that trend with your infrastructure products. Maybe before we get into those products, how has AI adoption kind of progressed inside AMD over the past several years? From a corporate perspective, what areas have seen the biggest productivity gains and what lessons from AMD's own AI journey are applicable to enterprise customers today?

DAN MCNAMARA: I can--

MATT RAMSAY: You want start?

DAN MCNAMARA: I can jump with that, yeah. So look, it's a great question. And I think that when I think about our journey, it's very similar to a number of enterprises. But our team started out, I think it's a multi-layer approach to the infrastructure.

We started out, first and foremost, with the data layer and optimized that. And we talked to a lot of enterprise customers and this is often overlooked, is how do you structure your data such that you can actually employ agents effectively? And we actually open sourced our solution. It's called Optima.

So we started there and then we've been on this journey about for agents driving what I would call automation for efficiency and that's gone very, very well. And now, we're at where, I would say, is we're really in the domain-specific type applications. So if you think about it, for us, domain specific is EDA.

So we're seeing a tremendous amount of upside across coding, debug, and those two key areas, along with kernel development, just software development in general, very, very strong returns there. And then of course, across all of the businesses, we're seeing very, very strong automation and efficiencies across each of the businesses. So I would say that, and it's interesting, because we were in New York City last week and I was with our CIO, and we had a roundtable with a number of top enterprise customers in New York City.

And he started out and just walked them through the journey. And it was a very good conversation about where each one of them are on this journey. So I would say that we're advanced in this area.

I would say that we took it on very, very aggressively. And we're also looking at, how do you balance token costs with the value? And we're really doing some advanced things across that too. So overall, very, very strong adoption. You got to look at this as both a provider and a major adopter of AI.

MATT RAMSAY: Yeah. Jim, the only thing I would add there is, you guys saw us work with a big framework that we put together with Anthropic about, obviously, them buying up to 2 gigawatts worth of MI450. But there's also a lot of work of not just the OpenAI tools, but the Anthropic Claude tools being adopted across our engineering organizations and unlocking a much faster flywheel of software development, debug, time to production of CHIP programs, optimizing where our software people are spending their time.

We have a huge software organization and trying to figure out what they need to be working on. Where can they use tools to accelerate that flywheel versus doing anything manual? I mean, my boss, Jean our CFO, has benchmarked us versus a whole bunch of leading semis and tech companies. And I think we're on the bleeding edge of AI adoption internally and it's come with an increased token cost, but it's come with a much, much greater productivity gain across the organization. And I think you'll see it allow us to bring hardware and software products to market much more quickly as we go forward.

DAN MCNAMARA: Yeah. Actually, just one last point. I want to just emphasize that, right? So you've got domain-specific and then you've got what I would call general IT automation.

And one's for efficiency, but when you can drive a faster time to market, that's where the real rubber hits the road. And that's what we're after. As we go to external enterprises, our goal is to get them time to value very quickly, with ROCm and with some of our solutions. So it's, again, we always say, we eat our own dog food. Everything we build is deployed in our data centers first and it's going very well in terms of driving our time to market with our engineering teams.

JIM SCHNEIDER: Yeah. And then with respect to your customers, from their perspective, how do you think this plays out in terms of model evolution over three to five years, in terms of the landscape? I mean, do you think frontier models are still going to be leading the charge here? Do we see small language models do a lot more of task-specific things? Or do you think open weight, open source models are going to have a larger role to play?

DAN MCNAMARA: You want to start or I can?

MATT RAMSAY: Yeah. I think, Jim, the answer is yes. It's not a very helpful answer, but it actually is the answer. I mean, our goal is to make sure that our combination of CPU and GPU roadmaps are very differentiated in terms of driving tokens per dollar outcomes, regardless of whether it's open weight models or frontier models for our largest customers.

I think those, I mean, obviously, the industry is evolving quickly around, what are the right use cases? How should we say this? How to apply the right tokens to the right problem relative to the cost of the token versus the return of the token? And that's a very large continuum.

I think our goal is to make sure that on the GPU side, our hardware and software are driving the right efficiencies, regardless of whether it's open weight models or our closed weight models or frontier models. And that Dan's business is the right CPU to run agents to drive all of those models, regardless of where they come from. I mean, that's kind of our goal. I don't know, Dan, if you--

DAN MCNAMARA: Yeah, I would just say, look, we rolled this out. I showed this at our Advancing AI Day. And Matt's right. It's all of the above, right?

Clearly, frontier will continue to be the cutting edge, but open weights are very, very valuable. And then you've got what I would call SLMs for some of this domain-specific stuff. And what we showed was intelligent routing.

So if you think about it, you've got frontier. Every enterprise is going to have some distributed model around frontier, probably GPU as a service in the cloud, most likely an on prem server that can service and run open weights models. And you have a policy-based router, depending on the task. And you're looking at performance latency, you're looking at, obviously, security.

That's one of the key areas where, what I hear mostly is cost and security from the enterprise. And what you can do is you route this and you can manage your costs. You can manage, if it's a policy-based router, if it's highly secure, it stays on prem. So we see a lot of enterprises trying to build this out. And it's very interesting because enterprises are infinitely hybrid, and we believe that will continue.

JIM SCHNEIDER: Great. Now we're going to dive straight into your business. First, the AI business and also the server CPU business as well. Your AI data center business has grown very rapidly over the last few years. If you think about the biggest strides you've made in product development across silicon software, customers, ecosystem, where do you think you can make the biggest strides going forward and what are your key focus areas from here?

DAN MCNAMARA: Yeah. This is a great question because, first and foremost, and I always say this because it's very, very important. Our vision for many years now has been, you build the right compute engine for the right workload. And that's across CPUs.

That's within not only across the product lines, but within the product lines. And we'll talk about server at some point, but we optimize for workloads. But most importantly is, we feel like we're in very, very good shape across the different product lines, right?

From server to GPU to networking. And now ROCm's coming online. So I think the biggest part for us is we have now shifted from this individual product lines to a full system provider. So providing the full rack, all of it interworking. And we're also driving a different roadmap cycle.

It used to be, three, five years ago, it was like, you're optimizing for your product. Now, it's a combined data center roadmap steering group. So whatever I'm doing, you have to make trade offs across all the products. So I think that's the biggest change. And what you'll see is getting rack scale solutions at scale is the biggest thing we're focused on right now.

MATT RAMSAY: I think from my perspective, just listening to Dan spoke about it just now, but listening to Lisa and others speak about, we don't necessarily have to force ourselves to be-- If you step back and think about the top, there's a long tail of customers that we're going to continue to support. But if you think about the large top 15 or 20 or consumers of high performance computing cycles in the world, we don't need to necessarily be their CPU partner, or GPU partner, or FPGA partner, or semi-custom partner. We can walk into a room strategically and say, How can we, at scale, be your high performance computing partner?

And that might look differently at different customers, but it's a very powerful thing to be able to say, Hey, we just want to be your high performance computing partner, and let's think strategically about what you want to do over the next number of generations and put solutions together that can support that across endpoint, across inference at the edge, across the server, on prem and in the cloud, AI deployments, in massive data center scale or in PCs. There's a huge continuum of, how can we be your high performance computing partner? And being able to bring those pieces of IP to the market at significant scale is one of the things that, I think, is unique about what we bring, is it's not a push approach.

It's a, how can we be your partner and let's decide how we're going to work together to bring significant amounts of high performance computing to market. I think that's the biggest change that's happened. And now that AMD has the full breadth of portfolio and the scale that we have, that's a conversation that I think is valuable.

JIM SCHNEIDER: Great. Now, the company has outlined some pretty healthy revenue growth targets, 60% CAGR over the next several years in data center revenue. 80% CAGR in AI data center revenue. Talk about two elements of that. One is, how diverse does this get between the hyperscalers, CSPs, enterprise, AI labs over time, even sovereigns? So how diverse does it get? And then secondly, what should we be thinking about in terms of markers for more of the short term going into 2027?

MATT RAMSAY: I'll start? Yeah, maybe I'll start and Dan can add a bunch of detail on his business and server. Yeah, Jim, we have outlined and we started at the Analyst Day back in November, and it's amazing how long ago that seems, given how fast this industry is moving now. But we talked about more than 60% growth of the data center franchise, more than 80% of growth of the AI business.

And at that time, we thought we were well above where the market was in talking about a $60 billion server TAM, and we've now more than tripled that. And so we're, at that point in time, talked about the company growing at more than 35% annually. Lisa and the team have updated the TAM for AMD to be more than around $2 trillion by 2030 and that's a 40% growth rate of the TAM.

And we expect to grow faster than that as a company. And we talked about getting to more than $20 in earnings over the strategic time frame. And I think we've updated that to be significantly more than $20. So we're excited about the growth, the leverage in the model.

And we've given a few data points on 2027, significantly, much more than doubling the data center business, and those are things that we feel really good about. And now it's just putting our heads down and making sure that we scale the AI business in terms of building racks. And I mean, Dan's business is in a very, very different place than it was 12 or 24 months ago in terms of growth.

So we feel it's a very, very exciting time for the company. But at the same time, we're heads down and trying to execute. So, Dan, if you want to expand on that.

DAN MCNAMARA: Yeah. I would just say, look, the way I look at our AI business is very similar to the way I looked at the server business five years ago, right? Is, very deliberate approach. You get in, and if you look at what we did in server, it was strong in cloud and national labs and then we evolved into the enterprise. And now we're seeing very, very strong growth in the enterprise.

I think you'll see the same thing happen. Like Matt said, we're very focused on delivering to our top customers right now with Helios, But. The spread will happen. Just like I just talked about. The enterprises are really thinking through what their overall infrastructure needs to look like.

It will include cloud, but if you think about AI, it's the exact opposite of what happened in general purpose compute. General purpose compute started on prem and went to the cloud. It's the exact opposite.

And we are seeing many of the mainstream enterprises look at building subrack scale, whether it's PCI card type of deployments or 8-way server, UBB-based deployments, to do exactly what I just talked about in terms of, what is the right balance and what's the distributed architecture that you need for the long term? So I think what you'll see is the shift happen over time. But right now, like Matt said, we're concentrated from an AI standpoint.

However, with server, Lisa and Jean talked about the results we're seeing across the enterprise, as well as cloud, and it's growing quite dramatically right now in terms share gains across all of the mainstream enterprise and the channel. We've invested very, very heavily over the last few years to go drive the channel and the enterprise, and it's really starting to pay off. So there's no fixed ratio, but it's more of, I see the same evolution happening across the AI business.

JIM SCHNEIDER: Fantastic. I want to dive into server CPUs next. Your home turf, so to speak. So for investors less familiar with the technical details of agentic AI, maybe help us understand why agentic workloads actually drive higher attach rates for CPUs? And as you do that, maybe talk about the changes in system architectures that occur as customers move from simple inference to more autonomous, multi-step AI workflows.

DAN MCNAMARA: Yeah. Look, this is a hot topic. And I think I would start with saying that this is more of a distributed systems architecture problem as opposed to a simple linear problem. If you think about the world of ChatGPT from November '22 to probably into last year, very linear. It was a SaaS-based data center. You

Have your servers for web serving. You got your application servers. You've got your database storage. You've got caching, and then you've got this GPU server.

Which, everyone understands a GPU server, right? You know the ratios. Everyone can calculate that very easily. And that was very linear, prompt response.

That's what it was built for. Well with agentic, as you all know, it's an entirely continuous flow. It's a completely different compute paradigm. It's 24/7, churning, within a sandbox spawning numbers of different agents.

So you just think of the picture I tried to just draw for you, if you think of your traditional servers here and your big GPU servers here, you kind of open it up and you pull in a whole new class of compute, which is for agentic control plane, API calls, database queries, tool execution. And that is pure CPU based. So that clearly will do RL with the GPU servers. So the GPU servers grow also.

But if you think about those general purpose servers, those get uplifted too, because more and more calls to those. So you're seeing an uplift in a whole new class plus the traditional general purpose. And we're just seeing that dramatically grow. At our FAD in November I said that, look, there's multiple areas of growth for the CPU.

We called it, but we called it too low. So we've upped it now. And I think the growth we're seeing across both the enterprise and the cloud is very, very exciting. And then lastly, what I would say is, with Venice, we are hitting on the three main focus areas for CPU.

You've got your GPU server that everyone knows and loves, in terms of started out 1 to 4, a CPU to GPU. Then you've got this agentic sandbox CPU where, with Venice, with our high core count, 256-core device, if you think about agentic, it is really threads per watt with the right level of per Core Performance. If you think about the head node, it's really about IPC and high frequency driving and keeping the GPUs busy. And then the general purpose servers, we've been very, very strong there for many years and we're going to continue.

So when you think about it, we feel like with not only with Turin today, leadership, Venice, we launched it already, and as it comes online here through the back half of this year, we are extremely well positioned to capture this growth. But I'd say one last thing. If you're trying to find a number to plug into a model, it's very, very hard because there's so many things.

If you just think of a gigawatt of power, and then you factor in your PUE and you come up with your IT power. It's all about the addition of the CPUs. Again, the host node, you know, we all know, that's easy calculation, but it all depends on what you're trying to run. It's really workload dependent and that's why it's so hard to plug a number in.

But that's why we tried to capture, Hey, this is the growth we see. And when we show it for agentic, it is also pulling in the uplift in those general purpose servers that I talked about. So I don't know if I confused you more or not, but I'm just trying to give you the picture of what we're seeing.

MATT RAMSAY: Yeah. And Dan, maybe I just add one thing. I mean, we did take a $60 billion TAM out to 2030 and up that now to 120 billion, and then 220 billion. And the companies, I know what Lisa's expecting of you, Dan, is for your business to be over 50% of that TAM as we grow.

And it'll be, we can do a relatively small number of chiplets and put them together in configurations that can be a significant number of SKUs and a full coverage of the platform. So, I mean, you guys can do the math on more than 50% of the 220 billion. I mean, I've been following and now part of AMD's server business for a very, very long time.

And to talk about building $100 billion server business is pretty exciting. No pressure, Dan. But that's what we see coming is a significant growth of agentic sandbox CPUs, for which we have very large core count, multithreaded parts of strong growth of head node CPUs, where we have really high frequency focused, high bandwidth, high single-threaded performance parts, and then the broad range of the server market.

One of the things that stuck out to me, seeing the results of Dan's business, is in the second quarter-- I mean, it seems like forever we talked about the second quarter, but that even the enterprise part of the server business grew more than 70%. The industry is not seeing those type of growth rates in enterprise server, basically, ever. So we're very excited about all parts of the server business, and the breadth of SKUs and the breadth of platforms as we roll out Venice and then move into the Florence generation is something that we're really excited about.

JIM SCHNEIDER: Yeah. Now, the server CPU market, as they said, as you talked about, it also becoming increasingly competitive, even as it's growing. So what advantages do you think the X86 ecosystem continues to provide for the enterprise, specifically? And how do you think about the durability of X86 in the hyperscaler environments, especially for some of these internal workloads where customers are developing their own silicon?

DAN MCNAMARA: Yeah. This is a common question. So I mean, first and foremost, we always talk about this, right? This is not an instruction set architecture problem or concern.

There's no fundamental differences in the ISA between X86 and ARM. It's really about delivering to different optimization points, right? It's perf per watt per dollar, ultimately. And that if we continue to drive along the three swim lanes that we just talked about and optimize for that performance per watt, we're in very, very good position.

And if you think about, from an ecosystem standpoint, if you go back to that picture, I tried to draw with my hands, all those general purpose servers that I talked about, X86-based today. Lots of software built for X86. So the ecosystem is built around X86.

So all that growth comes on X86. Now, if you look at the hyperscalers, each one of them are doing some form of their own. And what we see is, if we continue to drive, just what I talked about, which is the highest throughput and core density per watt, and then we hit these other points, we feel extremely good about the design in that we have right now across all of the major cloud vendors in the world.

From an agentic standpoint at 256 core, from a high frequency standpoint at 96 core, and then just across other SKUs for high performance computing. And even though, I'll just give you a good example. Recently, Amazon came out with RDS, which is their database service, which is a first party property that we would classify. It's on Turin and the reason why is performance.

So we just know that, yes, there is a focus for them to try and get their first party properties on their homegrown. But it doesn't fit for everything. And again, even when you go high density, it's that perf per core sweet spot and that optimization point on the VF curve that we pay close attention to.

So we really feel like where we are today, with coming up with Venice. Well, Turin today with Venice coming out as we speak and ramping. And then I was looking at we had a review earlier this week on even Zen 8 in terms of what our engineering teams are targeting.

So I feel very, very good about where we are in terms of delivering the optimization points. That's the key. It's really optimizing for the different workload and the deployment model.

MATT RAMSAY: I think, Dan, I agree. From my perspective, it's not watching the teams internally. The investor focus tends to be much more around instruction set. And it is important for the enterprise pieces of the server market, whether that's on prem deployment or in cloud deployment, but the economics of rolling out the server market to so unprecedented scale that we talked about with the TAM.

It's about building the best server parts, period. Never mind the instruction set. And I think that's what we, from a scale and supply chain point of view, from a-- I think you said optimization points and the number of SKUs that we can roll out, the number of platforms that we can roll out, the significant amount of optimization you can do for different places in the roadmap, I feel really good about where we are.

It's not just a, what we think about the market. We can see the demand pull from customers for different optimization points. And so when we think about, OK, this is where the demand pull is and these are conversations that are multi-generation in nature, I think we feel really good about where the server business is.

DAN MCNAMARA: I would just, final point on that is, for Venice, and I'm pretty sure Lisa talked about this at our last earnings, but with each generation we build, builds on the next. And you get more and more of the ecosystem coming along with you as you go and we've been very, very focused on that.

But with Venice, it's the broadest, true launch that we've had in terms of OEMs, ODMs, cloud vendors, the ISVs on day 0 support. The demand is very, very strong due to the three swim lanes that I talked about. I think our customers and ecosystem are seeing that one SKU doesn't solve every problem and that's what we're seeing from a merchant arm standpoint. It's really just singular SKUs or one or two SKUs. So we're pretty excited, because it really is, we're in a very good spot from a market opportunity standpoint and our product portfolio leadership across, really, I would argue, three generations straight.

JIM SCHNEIDER: Excellent. One of the things that's striking me over the past couple of years, we've kind of changed the parlance of how we talk about this market. We're not talking about server counts or counting accelerators.

We're talking about counting gigawatts of capacity. And every single presentation at this conference has done that. So maybe, as you think about these multi-gigawatt AI deployments, how should investors be thinking about CPU content per gigawatt?

DAN MCNAMARA: Do you want to?

MATT RAMSAY: Yeah, maybe I'll start. We think the focus that we have at AMD, broadly, in our data center business is to make sure that we provide very compelling tokens per dollar and TCO for our GPU business. And we're right in the thralls of ramping and launching Helios and MI455. And you'll see us be a very large partner to some of the leading model companies in the world to run their inference workloads.

Separately, Jim, regardless of whether the inference runs on our GPUs, or NVIDIA GPUs or TPUs or whatever, XPU, I mean, Dan can expand on this, but I think what we're focused on in the server business is to make sure that AMD's Venice portfolio and going forward, they are differentiated and right place for the industry to run agent code. And so we haven't been super specific about what that ratio is in terms of gigawatts of deployment, because it does look different depending on what customer it is. But we want to grow a very large AI business. And I think Dan's business is positioned to be a significant majority of the industry running agents to power agentic AI. And so we haven't been super specific on the gigawatt comments in terms of CPU.

DAN MCNAMARA: Yeah, I was just really simple, it depends. Because the challenge is, so take a gigawatt. Again, do your PUE.

You've got this IT and then you've got to break it down where, OK, I've got clusters of GPUs over here training. I've got clusters here doing inference. Then I've got to support it with a general purpose and then the agents, and it just really depends on what you're trying to accomplish with that gigawatt.

And it's very hard to just say, Oh, here's a fixed ratio. I would say that it's growing, right? If you think about it today, we're saying 1 to 1 ratio and it's going to continue to grow. But it's just very hard to pinpoint, plug this into a model and you'll get what you're looking for. It's very highly dependent on what the end customer is trying to accomplish.

JIM SCHNEIDER: I spent a lot of time plugging numbers into models, so, I get it. OK, we're almost out of time, but let me just leave you with the last question for you. We've covered a lot of ground. If you think about your position in AI, compute, data infrastructure, data center infrastructure, et cetera, if we're up here on stage again in five years and we look back, what do you think the one thing or two things that investors are going to be most surprised about in terms of the performance of the company?

MATT RAMSAY: Dan, do you want to start off?

DAN MCNAMARA: Look, I think, maybe I'll start with maybe what people may be missing about us. And it's what I talked about earlier, is we have fully transitioned from a very, very good silicon provider across multiple products, and we are transitioning now to full rack scale. And our software has come, even over the last six months, the gains we've seen-- We're becoming more of a software company and a systems company today than we were even six months ago. So I would just say that will be, I think, if you look forward 12 to 24 months, I think it will become very clear how we have made that transition quickly and we've driven a software stack that is truly focused on time to value for our customers. I think that's where I would leave it in terms of what you'll see over the next few years.

MATT RAMSAY: I mean, from my perspective, the goal is to-- I mean, Dan started the conversation this way, Jim, where we want to provide the industry that consumes high performance computing with the right type of computing for the right type of workload. And I think that, that will serve us well across the breadth of our markets. And we're right now, at one of the more exciting times that the industry has seen and more exciting times for the company.

We've talked about much more than doubling our data center business next year, and driving gross margin dollars very significantly faster than expenses. So we're at that inflection point and as I think it's important for the investor community to understand that Lisa and the whole team-- It was funny. We were doing a meeting in a room at the conference just an hour before we came on stage here and Dan was on an execution meeting with Lisa and the team.

So it's like, the team is focused on making sure that we have a cadence of execution at the company. And despite all the excitement there, the focus remains on making sure that we execute. And if we do that, then I think investors will be really pleased with where things end up. For us, it's about driving outcomes for customers and then that will translate into outcomes for the investment community, not the other way around. So we're just going to put our heads down and execute because it's a super exciting time.

But as we wrap up here, the little blinky light is on, but thank you all for Spending time with us. And thank you, Jim, and the team at Goldman for hosting us. We really appreciate it.

JIM SCHNEIDER: And Matt, thanks for being here.

MATT RAMSAY: All right. Appreciate it.

JIM SCHNEIDER: Thank you.

DAN MCNAMARA: Thank you.